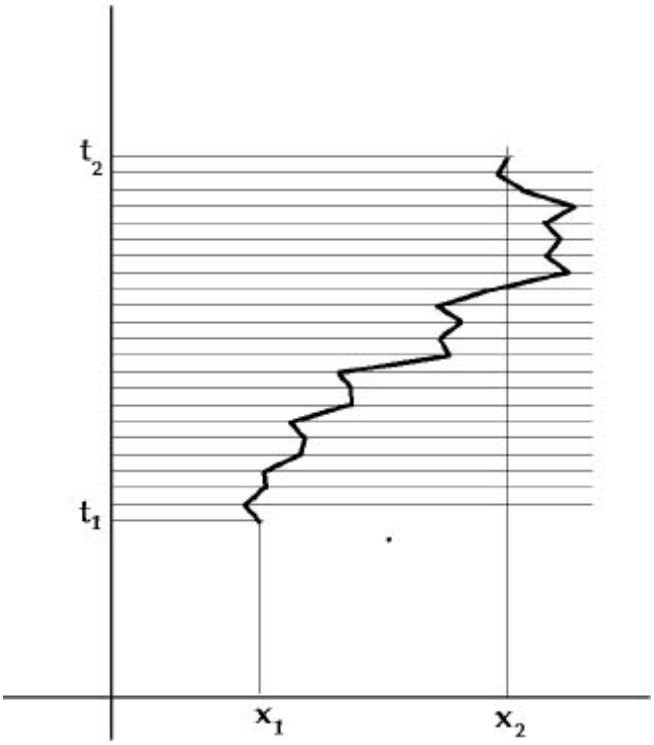


Figure 9.7: Further Steps Toward Constructing the Path Integral. Keeping the same starting and ending points, break the path up into a large number of equally sized segments.

Now, let's turn to quantum mechanics. The idea of a well-defined trajectory between two points makes no sense in quantum mechanics because of the uncertainty principle. However, a question that we *can* ask is: Given that a particle starts out at (x_1, t_1), what is the probability that it will show up at (x_2, t_2) if an observation of its position is made?

As always in quantum mechanics, the probability is the square of the absolute value of a complex amplitude. The global version of quantum mechanics asks:

> *Given that a particle starts out at* (x_1, t_1)*, what is the amplitude that it will show up at* (x_2, t_2)*?*

Let's call that amplitude $C(x_1, t_1; x_2, t_2)$ or, more simply, just $C_{1,2}$. The initial state of the particle is $|\Psi(t_1)\rangle = |x_1\rangle$. Over the time interval between t_1 and t_2, the state evolves to[10]

$$|\Psi(t_2)\rangle = e^{-iH(t_2-t_1)}|x_1\rangle. \qquad (9.20)$$